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Kendall & Kendall
Systems Analysis and Design, 9e

Understanding and Modeling Organizational Systems

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Entity-Relationship Model

- Focus is on the entities and their relationships within the organizational system
- Another way to show the scope of a system

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Relationships

- Relationships show how the entities are connected
- Three types of relationships:
  - One-to-one
  - One-to-many
  - Many-to-many

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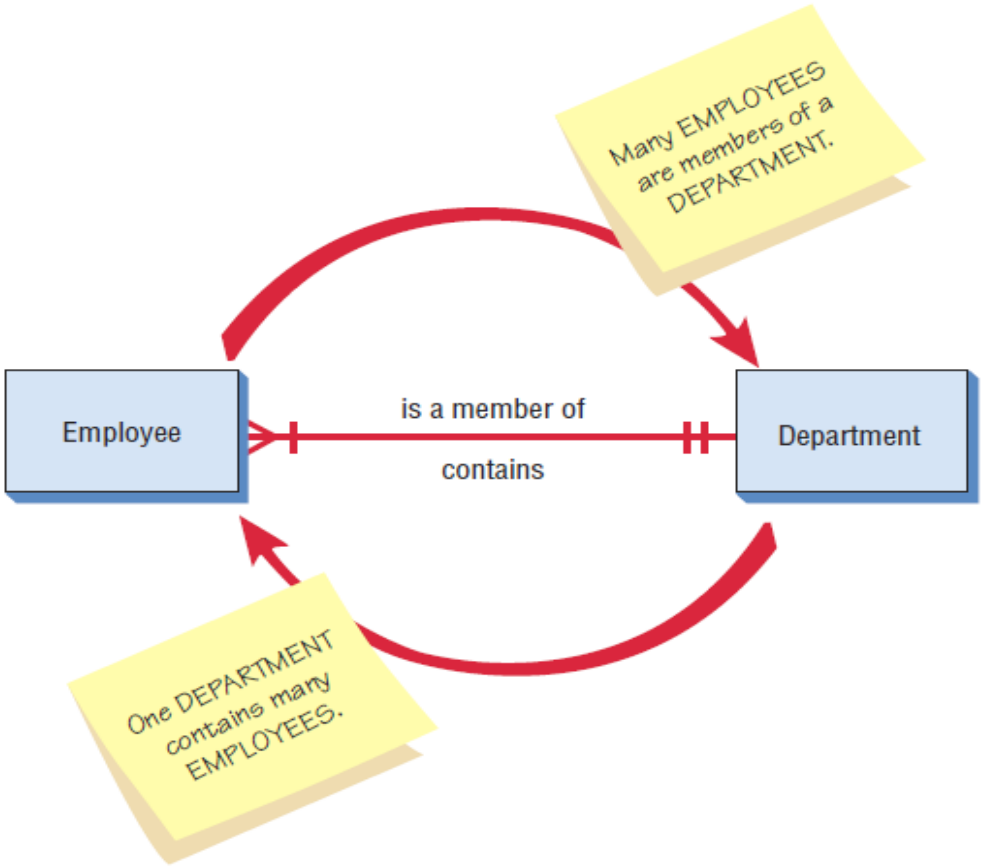
Entity-Relationship Example (Figure 2.7)



An entity-relationship diagram showing a many-to-one relationship

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Creating Entity-Relationship Diagrams

- List the entities in the organization
- Choose key entities to narrow the scope of the problem
- Identify what the primary entity should be
- Confirm the results of the above through data gathering

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Use Case Modeling

- Describes what a system does without describing how the system does
  - A logical model of the system
- Use case is a view of the system requirements
- Analyst works with business experts to develop requirements

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Use Case Diagram

- Definition:
- A diagram that shows a set of use cases and actors and
- their relationships.
- Use cases represent system functionality, the requirements
- of the system from the user's perspective.

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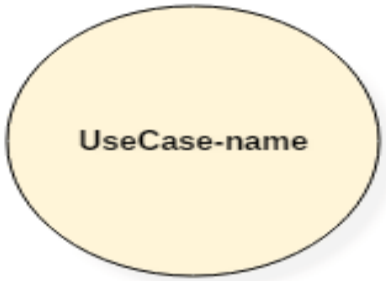
Use-case diagram notations

Following are the common notations used in a use case diagram:
Use-case:
Use cases are used to represent high-level functionalities and how the user will handle the system. A use case represents a distinct functionality of a system, a component, a package, or a class. It is denoted by an oval shape with the name of a use case written inside the oval shape. The notation of a use case in UML is given below:



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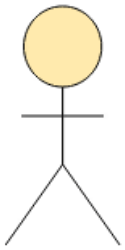
Actor:
It is used inside use case diagrams. The actor is an entity that interacts with the system. A user is the best example of an actor. An actor is an entity that initiates the use case from outside the scope of a use case. It can be any element that can trigger an interaction with the use case. One actor can be associated with multiple use cases in the system. The actor notation in UML is given below.



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How to draw a use-case diagram?

- To draw a use case diagram in UML first one need to analyse the entire system carefully. You have to find out every single function that is provided by the system. After all the functionalities of a system are found out, then these functionalities are converted into various use cases which will be used in the use case diagram.
- A use case is nothing but a core functionality of any working system. After organizing the use cases, we have to enlist the various actors or things that are going to interact with the system. These actors are responsible for invoking the functionality of a system. Actors can be a person or a thing. It can also be a private entity of a system. These actors must be relevant to the functionality or a system they are interacting with.

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Use Case Relations

- Behavioral relationships
  - Communicates
    - Used to connect an actor to a use case
  - Includes
    - Describes the situation in which a use case contains behavior that is common to more than one use case

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Use Case Relations

- Behavioral relationships (continued)
  - Extends
    - Describes the situation in which one use case possesses the behavior that allows the new case to handle a variation or exception from the basic use case
  - Generalizes
    - Implies that one thing is more typical than the other thing

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Four Types Of Behavioral Relationships And The Lines Used To Diagram Each (Figure 2.13)

Relationship	Symbol	Meaning
Communicates		An actor is connected to a use case using a line with no arrowheads.
Includes	<< include >>	A use case contains a behavior that is common to more than one other use case. The arrow points to the common use case.
Extends	<< extend >>	A different use case handles exceptions from the basic use case. The arrow points from the extended to the basic use case.
Generalizes		One UML "thing" is more general than another "thing." The arrow points to the general "thing."

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Some components of use case diagrams showing actors, use cases, and relationships for a student enrollment example (Figure 2.14)

- After the actors and use cases are enlisted, then you have to explore the relationship of a particular actor with the use case or a system. One must identify the total number of ways an actor could interact with the system. A single actor can interact with multiple use cases at the same time, or it can interact with numerous use cases simultaneously.

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Following rules must be followed while drawing use-case for any system:

- The name of an actor or a use case must be meaningful and relevant to the system.
- Interaction of an actor with the use case must be defined clearly and in an understandable way.
- Annotations must be used wherever they are required.
- If a use case or an actor has multiple relationships, then only significant interactions must be displayed.

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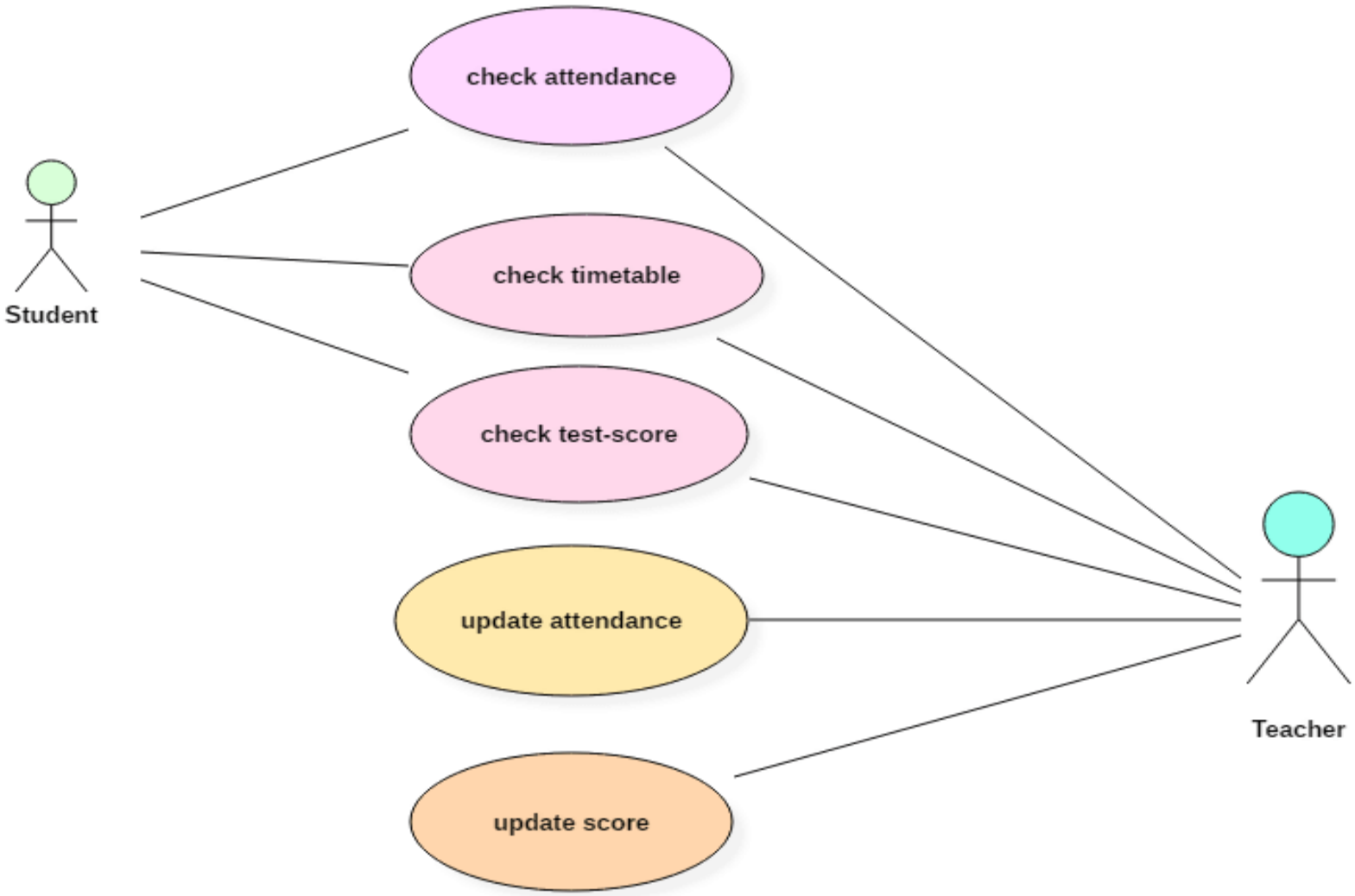
An example of a use-case diagram

Following use case diagram represents the working of the student management system



- In the above use case diagram, there are two actors named student and a teacher. There are a total of five use cases that represent the specific functionality of a student management system. Each actor interacts with a particular use case.
- A student actor can check attendance, timetable as well as test marks on the application or a system. This actor can perform only these interactions with the system even though other use cases are remaining in the system.

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- It is not necessary that each actor should interact with all the use cases, but it can happen.
- The second actor named teacher can interact with all the functionalities or use cases of the system. This actor can also update the attendance of a student and marks of the student. These interactions of both student and a teacher actor together sums up the entire student management application.
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- When to use a use-case diagram?
- A use case is a unique functionality of a system which is accomplished by a user. A purpose of use case diagram is to capture core functionalities of a system and visualize the interactions of various things called as actors with the use case. This is the general use of a use case diagram.
- The use case diagrams represent the core parts of a system and the workflow between them. In use case, implementation details are hidden from the external use only the event flow is represented.
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In general use case diagrams are used for:

- Analyzing the requirements of a system
- High-level visual software designing
- Capturing the functionalities of a system
- Modeling the basic idea behind the system
- Use cases are intended to convey desired functionality so the exact scope of a use case may vary according to the system and the purpose of creating UML model.

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Summary

- Use case diagrams are a way to capture the system's functionality and requirements in UML diagrams.
- It captures the dynamic behavior of a live system.
- A use case diagram consists of a use case and an actor.
- A use case represents a distinct functionality of a system, a component, a package, or a class.
- An actor is an entity that initiates the use case from outside the scope of a use case.
- The name of an actor or a use case must be meaningful and relevant to the system.
- A purpose of use case diagram is to capture the core functionalities of a system.

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Draw a use case diagram for the scenario below: Inventory System

- In order to generate an invoice a clerk must log in. If a clerk is a first time user, one must have themselves registered.
- There should be an option for a user to register oneself within the login page.
- Any user can use the system to view products
- online.
- The option of login is also provided when a user views products online.

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UML

- UML short for Unified Modeling Language, is a standardized modeling language consisting of an integrated set of diagrams, developed to help system and software developers for specifying, visualizing, constructing, and documenting the artifacts of software systems, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems. The UML is a very important part of developing object oriented software and the software development process.

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UML

- The UML uses mostly graphical notations to express the design of software projects. Using the UML helps project teams communicate, explore potential designs, and validate the architectural design of the software.

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Draw A Use Case Diagram

- Web Customer actor uses some web site to make purchases online. Top level use cases are View Items, Make Purchase and Client Register. View Items use case could be used by customer as top level use case if customer only wants to find and see some products. This use case could also be used as a part of Make Purchase use case. Client Register use case allows customer to register on the web site, for example to get some coupons or be invited to private sales. Note, that Checkout use case is included use case not available by itself - checkout is part of making purchase.

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